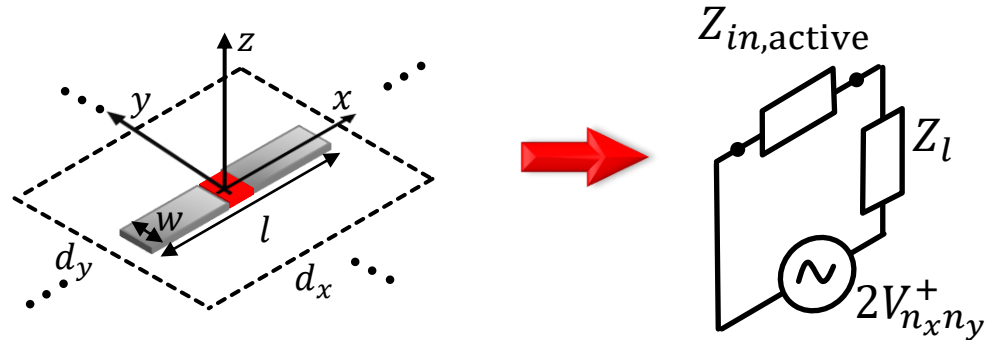


EE4725 Quasi Optical Systems

Matlab exercise: Active input impedance

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Let us consider a doubly-periodic infinite array of dipoles in free space



The active input impedance for a dipole array in free space can be expressed as

$$Z_{in,active} = -\frac{1}{d_x d_y} \sum_{m_x=-\infty}^{\infty} \sum_{m_y=-\infty}^{\infty} G_{xx,FS}^{ej}(k_{xm}, k_{ym}) |I(k_{xm})|^2 |J_t(k_{ym})|^2$$

where

$$I(k_x) = \frac{2k_0 \left(\cos\left(\frac{k_x l}{2}\right) - \cos\left(\frac{k_0 l}{2}\right) \right)}{(k_0^2 - k_x^2) \sin\left(k_0 \frac{l}{2}\right)}, \quad J_t(k_y) = \text{sinc}\left(\frac{k_y w}{2}\right)$$

As an example, the geometrical parameters are

$$\begin{aligned} d_x &= 15e-3; \\ d_y &= 15e-3; \end{aligned}$$

$$\begin{aligned} w &= 3e-3; \\ l &= 13.5e-3; \end{aligned}$$

1) Implement the expression of the active input impedance in Matlab and plot the variation of real and imaginary part with the frequency (in the range 5-15 GHz) for

- Case 1

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Theta = 0;  
Phi    = 0;
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- Case 2

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Theta = 30*pi/180;  
Phi    = 0;
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2) Plot also the variation with frequency of the

- Active reflection coefficient normalized to a proper Z_L
- Fundamental mode of $z_{in,active}$: Z_{00}
- Higher order modes of $z_{in,active}$: Z_{ho}

3) Are these quantities complex, real or imaginary and why?

4) At what frequency do you expect to find a grating lobe for Case 2? How do the impedance curves change at this frequency?
